Saraangan B

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.84
Sri Chaitanya Techno School, Hosur	Class XII (CBSE)	2024	93.60%
Sri Chatitanya Techno school, Hosur	Class X (CBSE)	2022	95.80%

PROFESSIONAL SUMMARY

Engineering Physics undergraduate at IIT Hyderabad with experience in LTSpice-based analog circuit simulation and Synopsys Sentaurus TCAD. Currently working on SPAD quenching circuit validation at DIRACQ Systems while pursuing interests in semiconductor devices, analog/mixed-signal circuits, and digital VLSI.

EXPERIENCE

DIRACQ Systems
Engineering Intern | Jan 2026 - Present
- Actively validated and modified a state-of-the-art quenching circuit for Single Photon Avalanche Diodes (SPADs) using LTSpice, performing transient analysis and circuit validation for quenching and recharge behaviour.
- Developed behavioural models for SPADs across various breakdown voltages and other circuit components.
- Assisted in component selection and verification.
- Assisted in technical documentation and product development.

PROJECTS

Synchronous & Asynchronous FIFO Designs in Verilog HDL (GitHub)

- Designed and implemented parameterizable synchronous and asynchronous FIFO architectures in Verilog HDL, incorporating Gray-code pointer synchronization, dual flip-flop synchronizers, and robust full/empty flag generation for safe clock-domain crossing.
- Developed comprehensive self-checking SystemVerilog verification environments featuring scoreboards, randomized stress testing, overflow/underflow validation, pointer wrap-around checks, and asynchronous clock scenarios.
- Generated an extreme testbench for verification stimuli and stress-test scenarios to improve test coverage, followed by manual debugging, refinement, and validation of the verification environment and RTL implementation.
- Verified functional correctness through extensive simulation, waveform analysis, and randomized verification using Icarus Verilog and GTKWave on EDAPlayground, documenting the complete design methodology in a public GitHub repository.

16× Oversampling Full-Duplex UART in Verilog HDL (GitHub)

- Designed and implemented a fully-duplex Universal Asynchronous Receiver and Transmitter (UART) in Verilog HDL with independent transmitter and receiver modules.
- Implemented a 16× oversampling receiver architecture, configurable baud-rate generation, and finite-state-machine based protocol handling for reliable UART communication across varying baud-rate conditions.
- Developed comprehensive Verilog testbenches to verify transmission, reception, and protocol functionality across multiple operating scenarios and validated the design through GTKWave waveform analysis and documented the architecture, implementation, and verification methodology in a public GitHub repository.

LTSpice Simulation Portfolio (Ongoing) (GitHub)

- Implemented analog circuit simulations for selected *SPICE Problems* from *Fundamentals of Microelectronics* by using LTSpice. Performed DC, transient, and AC simulations to analyze circuit behaviour and verify textbook exercises.
- Maintaining a public GitHub repository documenting circuit schematics, simulation plots, and annotated solutions.

SCHOLASTIC ACHIEVEMENTS

- Scored 94% of the total marks (96.45th percentile) in the **Level-2** of the **Indian Semiconductor Workforce Development Programme** conducted by the ***IISc Bangalore*** in collaboration with ***Synopsys***.

SKILLS

Languages: C, Python, Verilog HDL

Tools & EDA: LTSpice, Synopsys Sentaurus Workbench, Synopsys Sentaurus Device (SDevice), GTKWave

Core Domains: Analog Circuit Simulation, Digital Logic Design, Semiconductor Device Modeling

POSITIONS OF RESPONSIBILITY

- Core member of the Quiz Club of IIT Hyderabad (2025-2026)
- Head of the Quiz Club of IIT Hyderabad (2026-)

RELEVANT COURSES

Basic Electric Circuits | Analog Electronics | Digital Electronics | Electromagnetism | Modern Physics | Data Structures & Algorithms | Computational Physics | Introduction to Programming

EXTRACURRICULAR

- Selected to represent IIT Hyderabad in the Inter-IIT Quiz Competition.